



MTN CUSTOMER CHURN ANALYSIS DASHBOARD

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CHAPTER ONE

INTRODUCTION

1.1 Executive Summary

Customer retention remains one of the most significant strategic priorities in the telecommunications industry. As competition intensifies and customer expectations continue to evolve, telecommunications companies must leverage data-driven insights to understand customer behavior, improve service delivery, and minimize customer attrition. Customer churn not only reduces recurring revenue but also increases customer acquisition costs, weakens market share, and affects long-term profitability.

This project presents a comprehensive Business Intelligence solution developed using Microsoft Power BI to analyze customer churn within MTN. The analysis integrates customer demographics, revenue performance, subscription behavior, customer satisfaction, tenure, and churn patterns into an interactive dashboard environment designed to support executive decision-making.

The solution consists of five interconnected dashboards: Executive Overview, Customer Analysis, Revenue & Customer Value Analysis, Churn Deep Dive, and Customer Retention & Risk Analysis. Together, these dashboards provide management with a holistic view of business performance while identifying the major factors influencing customer churn and revenue generation.

The analysis revealed that although MTN generated substantial revenue from its customer base, approximately 29.44% of customers had churned, representing a considerable threat to future revenue growth and customer lifetime value. The findings further indicate that customer dissatisfaction, competitive market offers, service quality issues, pricing concerns, and regional performance disparities are among the leading drivers of customer attrition.

Based on these findings, this project proposes strategic recommendations including enhanced customer experience management, predictive churn analytics, personalized retention campaigns, improved network quality, optimized pricing strategies, and customer segmentation initiatives. These recommendations are intended to improve customer retention, protect recurring revenue, and strengthen MTN's competitive position within the Nigerian telecommunications industry.

Ultimately, this project demonstrates how Business Intelligence can transform operational data into meaningful insights that support evidence-based decision-making and sustainable business growth.

1.2 Business Background

The telecommunications industry plays a critical role in economic development by providing voice, data, and digital communication services to individuals and businesses. As one of Nigeria's leading telecommunications providers, MTN serves millions of customers across diverse demographic and geographic segments. However, the industry is characterized by intense competition, rapid technological advancements, evolving customer expectations, and increasing demand for reliable and affordable communication services.

In such a competitive environment, retaining existing customers is often more cost-effective than acquiring new ones. Research consistently shows that acquiring a new customer can cost several times more than retaining an existing one. Consequently, customer churn has become a key performance indicator that directly influences organizational profitability, customer lifetime value, and long-term sustainability.

Business Intelligence (BI) has emerged as a powerful approach for transforming large volumes of operational data into actionable insights. By integrating data from multiple sources, developing meaningful Key Performance Indicators (KPIs), and presenting findings through interactive dashboards, organizations can monitor performance, identify trends, evaluate customer behavior, and make informed strategic decisions.

This project applies Business Intelligence techniques to analyze MTN's customer churn dataset using Microsoft Power BI. Through data modeling, DAX calculations, Power Query transformations, and interactive visualizations, the project provides management with a comprehensive understanding of customer behavior, revenue trends, churn drivers, and retention opportunities.

The resulting dashboards are designed not only to monitor business performance but also to support proactive decision-making aimed at improving customer satisfaction, reducing churn, and maximizing organizational profitability.

1.3 Business Problem

Customer churn represents one of the most significant operational and financial challenges facing telecommunications companies. Every customer who leaves the network results in a direct loss of recurring revenue and often requires substantial marketing and promotional investments to acquire a replacement customer.

Despite generating significant revenue, organizations frequently struggle to identify the underlying causes of customer attrition due to fragmented data sources and limited visibility into customer behavior. Without comprehensive analytical tools, management may be unable to detect emerging churn patterns, identify high-risk customer segments, or implement timely interventions that reduce customer losses.

For MTN, customer churn may be influenced by several interconnected factors, including customer satisfaction, service quality, pricing, subscription preferences, tenure, network performance, and competitive market offerings. Understanding how these factors interact is essential for developing effective customer retention strategies.

Furthermore, executives require timely, accurate, and interactive reports that provide a holistic view of organizational performance rather than isolated operational metrics. Traditional reporting methods often fail to deliver the depth of analysis necessary for strategic decision-making.

Therefore, there is a need for an integrated Business Intelligence solution capable of analyzing customer demographics, revenue performance, customer behavior, churn drivers, and retention opportunities in a manner that supports evidence-based decision-making and improves organizational performance.

1.4 Project Objectives

General Objective

To develop a comprehensive Business Intelligence dashboard using Microsoft Power BI that analyzes customer churn, evaluates revenue performance, identifies customer behavior patterns, and supports strategic decision-making for improving customer retention and business performance.

Specific Objectives

The specific objectives of this project are to:

- 1) Develop an interactive Executive Dashboard that provides management with a comprehensive overview of key business performance indicators.
- 2) Analyze customer demographic characteristics, including age, gender, tenure, subscription plans, and geographical distribution, to understand customer behavior.
- 3) Evaluate revenue performance across products, devices, customer segments, and geographic locations to identify key revenue drivers.
- 4) Investigate the major factors contributing to customer churn and quantify their impact on organizational revenue.
- 5) Identify high-risk customer segments that require proactive retention strategies.
- 6) Measure customer satisfaction, retention rate, churn rate, customer lifetime value (CLV), and other key performance indicators.
- 7) Provide actionable business recommendations that support customer retention, revenue growth, and operational excellence.
- 8) Demonstrate the application of Business Intelligence tools, data modeling, DAX, and data visualization techniques in solving real-world business problems.

CHAPTER TWO

DATA PREPARATION AND MODEL DEVELOPMENT

2.1 Data Overview

Dataset Overview

The MTN Customer Churn dataset serves as the primary source of information for this Business Intelligence project. The dataset contains historical customer records designed to support the analysis of customer behavior, revenue performance, and churn patterns within a telecommunications environment. It provides valuable insights into customer demographics, subscription characteristics, service usage, customer satisfaction, and revenue generation, enabling a comprehensive assessment of the factors influencing customer retention and attrition.

The dataset comprises **496 customer records**, with each record representing an individual customer's interaction with the organization. The data captures both customer profile information and business performance indicators required for developing executive-level dashboards and key performance metrics.

The dataset includes several variables that describe customer demographics, subscription information, financial performance, and churn behavior. These variables form the foundation for calculating Key Performance Indicators (KPIs), performing customer segmentation, evaluating revenue performance, and identifying the drivers of customer churn. The major variables contained in the dataset include:

Variable	Description
Customer ID	Unique identifier assigned to each customer.
Gender	Gender classification of the customer.
Age	Customer's age used for demographic segmentation.
State	Customer's geographical location within Nigeria.
Subscription Plan	Service plan subscribed to by the customer.
Device	Type of device or service used by the customer.
Monthly Revenue	Revenue generated from each customer.
Customer Satisfaction	Customer satisfaction rating reflecting service experience.
Tenure	Length of time the customer has remained with MTN.
Churn Status	Indicates whether the customer has churned or remained active.
Churn Reason	Primary reason provided for customer attrition.
Purchase Date	Date associated with customer subscription or transaction activities.

These variables collectively provide sufficient information to analyze customer behavior, measure business performance, evaluate customer value, identify churn trends, and generate actionable business insights.

Figure 2.1: Raw MTN Customer Churn Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Customer	Full Name	Date of Pu	Age	State	MTN Device	Gender	Satisfactio	Customer	Customer	Subscripti	Unit Price	Number o	Total Reve	Data Usag	Customer	Reasons for Churn		
2	CUST0001	Ngozi Berry	Jan-25	27	Kwara	4G Router	Male	2	Fair	2	165GB Mo	35000	19	665000	44.48	Yes	Relocation		
3	CUST0002	Zainab Baker	Mar-25	16	Abuja (FC	Mobile SIM Card	Female	2	Fair	22	12.5GB Mc	5500	12	66000	19.79	Yes	Better Offers from Competitors		
4	CUST0003	Saidu Evans	Mar-25	21	Sokoto	5G Broadband Rout	Male	1	Poor	60	150GB FUf	20000	8	160000	9.64	No			
5	CUST0003	Saidu Evans	Mar-25	21	Sokoto	Mobile SIM Card	Male	1	Poor	60	1GB+1.5m	500	8	4000	197.05	No			
6	CUST0003	Saidu Evans	Mar-25	21	Sokoto	Broadband MiFi	Male	1	Poor	60	30GB Mon	9000	15	135000	76.34	No			
7	CUST0004	Ejiro Walker	Mar-25	36	Gombe	4G Router	Female	1	Poor	14	10GB+10m	4500	9	40500	92.72	No			
8	CUST0005	Nura Mann	Jan-25	57	Oyo	4G Router	Male	3	Good	53	25GB Mon	9000	16	144000	42.92	No			
9	CUST0006	Tamuno Lewis	Feb-25	74	Plateau	Mobile SIM Card	Female	5	Excellent	9	7GB Montl	3500	5	17500	44.25	No			
10	CUST0006	Tamuno Lewis	Feb-25	74	Plateau	5G Broadband Rout	Female	5	Excellent	9	1.5TB Year	150000	2	300000	18.64	No			
11	CUST0006	Tamuno Lewis	Feb-25	74	Plateau	4G Router	Female	5	Excellent	9	65GB Mon	16000	6	96000	30.99	No			
12	CUST0007	Boma Stewart	Mar-25	24	Jigawa	Broadband MiFi	Female	5	Excellent	39	120GB Mo	24000	11	264000	150.3	Yes	Relocation		
13	CUST0008	Ifearnyi Park	Jan-25	53	Kwara	Broadband MiFi	Male	3	Good	30	300GB FUf	30000	17	510000	168.41	No			
14	CUST0009	Bala Robbins	Jan-25	35	Gombe	Broadband MiFi	Male	5	Excellent	35	300GB FUf	30000	10	300000	175.81	No			
15	CUST0010	Halima Ochoa	Mar-25	23	Imo	4G Router	Male	1	Poor	6	10GB+10m	4500	20	90000	53.38	No			
16	CUST0010	Halima Ochoa	Mar-25	23	Imo	Broadband MiFi	Male	1	Poor	6	60GB Mon	14500	18	261000	192.89	No			
17	CUST0010	Halima Ochoa	Mar-25	23	Imo	Mobile SIM Card	Male	1	Poor	6	500MB Da	350	13	4550	142.73	No			
18	CUST0011	Ejiro Griffith	Feb-25	72	Bauchi	4G Router	Female	2	Fair	8	65GB Mon	16000	2	32000	14.66	Yes	Poor Network		
19	CUST0011	Ejiro Griffith	Feb-25	72	Bauchi	5G Broadband Rout	Female	2	Fair	8	60GB Mon	14500	2	29000	13.95	Yes	Poor Network		
20	CUST0011	Ejiro Griffith	Feb-25	72	Bauchi	Broadband MiFi	Female	2	Fair	8	300GB FUf	30000	8	240000	14.64	Yes	Poor Network		
21	CUST0012	Zainab Morton	Mar-25	78	Ondo	4G Router	Male	2	Fair	47	65GB Mon	16000	12	192000	131.97	No			
22	CUST0012	Zainab Morton	Mar-25	78	Ondo	Broadband MiFi	Male	2	Fair	47	60GB Mon	14500	2	29000	82.51	No			
23	CUST0013	Ngozi Moreno	Jan-25	23	Kebbi	4G Router	Female	2	Fair	52	10GB+10m	4500	8	36000	56.07	Yes	Relocation		
24	CUST0013	Ngozi Moreno	Jan-25	23	Kebbi	Mobile SIM Card	Female	2	Fair	52	3.2GB 2-D	1000	18	18000	88.9	Yes	Relocation		

2.2 Data Cleaning and Transformation

Data quality is fundamental to producing reliable analytical results. Prior to developing the dashboards, the dataset was subjected to a comprehensive data cleaning and transformation process using ****Power Query**** within Microsoft Power BI. This process ensured that the data was accurate, complete, consistent, and suitable for business analysis.

Several preprocessing activities were carried out to improve the quality of the dataset. These included validating data types, checking for missing values, reviewing duplicate records, and standardizing data formats across all variables.

The following data preparation activities were performed:

- Imported the MTN Customer Churn dataset into Microsoft Power BI.
- Verified the completeness and consistency of all customer records.
- Reviewed the dataset for missing or null values and ensured that critical analytical fields were complete.
- Confirmed that each customer was uniquely identified using the **Customer ID** field.
- Converted variables to their appropriate data types, including text, whole numbers, decimal numbers, and date formats.
- Standardized date fields to support accurate time-based analysis and DAX time intelligence calculations.
- Created additional calculated columns, including customer segmentation variables such as Age Group and Tenure Group, to facilitate demographic analysis.
- Developed a dedicated **Date Dimension** to support Month-over-Month (MoM) analysis, Year-to-Date (YTD) calculations, and other time intelligence functions.

- ix. Renamed selected columns to improve readability and maintain consistent naming conventions throughout the data model.

These transformation activities improved the overall integrity of the dataset and established a reliable foundation for subsequent analysis and dashboard development.

Figure 2.2: Power Query Data Cleaning and Transformation Process

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"Customer ID", type text}, {"Full Name", type text}, {"Date of Purchase", type date}, {"Age", type number}, {"State", type text}, {"MTN Device", type text})

	Customer ID	Full Name	Date of Purchase	Age	State	MTN Device
1	CUST0001	Ngozi Berry	01/01/2025	27	Kwara	4G Router
2	CUST0002	Zainab Baker	01/03/2025	16	Abuja (FCT)	Mobile SIM C
3	CUST0003	Saidu Evans	01/03/2025	21	Sokoto	5G Broadbar
4	CUST0003	Saidu Evans	01/03/2025	21	Sokoto	Mobile SIM C
5	CUST0003	Saidu Evans	01/03/2025	21	Sokoto	Broadband M
6	CUST0004	Ejiro Walker	01/03/2025	36	Gombe	4G Router
7	CUST0005	Nura Mann	01/01/2025	57	Oyo	4G Router
8	CUST0006	Tamuno Lewis	01/02/2025	74	Plateau	Mobile SIM C
9	CUST0006	Tamuno Lewis	01/02/2025	74	Plateau	5G Broadbar
10	CUST0006	Tamuno Lewis	01/02/2025	74	Plateau	4G Router
11	CUST0007	Boma Stewart	01/03/2025	24	Jigawa	Broadband M
12	CUST0008	Ifeanyi Park	01/01/2025	53	Kwara	Broadband M
13	CUST0009	Bala Robbins	01/01/2025	35	Gombe	Broadband M
14	CUST0010	Halima Ochoa	01/03/2025	23	Imo	4G Router
15	CUST0010	Halima Ochoa	01/03/2025	23	Imo	Broadband M
16	CUST0010	Halima Ochoa	01/03/2025	23	Imo	Mobile SIM C
17	CUST0011	Ejiro Griffith	01/02/2025	72	Bauchi	4G Router
18	CUST0011	Ejiro Griffith	01/02/2025	72	Bauchi	5G Broadbar
19	CUST0011	Ejiro Griffith	01/02/2025	72	Bauchi	Broadband M
20	CUST0012	Zainab Morton	01/03/2025	78	Ondo	4G Router
21	CUST0012	Zainab Morton	01/03/2025	78	Ondo	Broadband M
22	CUST0013	Ngozi Moreno	01/01/2025	23	Kebbi	4G Router
23	CUST0013	Ngozi Moreno	01/01/2025	23	Kebbi	Mobile SIM C
24	CUST0014	Ngozi Crawford	01/02/2025	22	Adamawa	4G Router
25	CUST0014	Ngozi Crawford	01/02/2025	22	Adamawa	Mobile SIM C

2.3 Data Modeling (Star Schema)

An efficient data model is essential for building scalable, high-performing Business Intelligence solutions. To improve analytical performance and simplify report development, this project adopted the **Star Schema** data modeling approach.

The Star Schema organizes data into a central **Fact Table** surrounded by multiple **Dimension Tables**, enabling efficient filtering, aggregation, and DAX calculations while minimizing redundancy.

The central **Fact Customer Churn** table contains the transactional and measurable business data, including customer revenue, churn status, customer satisfaction scores, subscription information, and purchase dates. This table serves as the primary source for calculating business metrics and Key Performance Indicators.

Supporting the fact table are several dimension tables that provide descriptive information used for slicing and filtering the data. The most important dimension created during this project is the **Date Dimension (DimDate)**, which supports advanced time intelligence calculations such as Month-over-Month Revenue Growth, Previous Month Revenue, Year-to-Date analysis, and trend visualization.

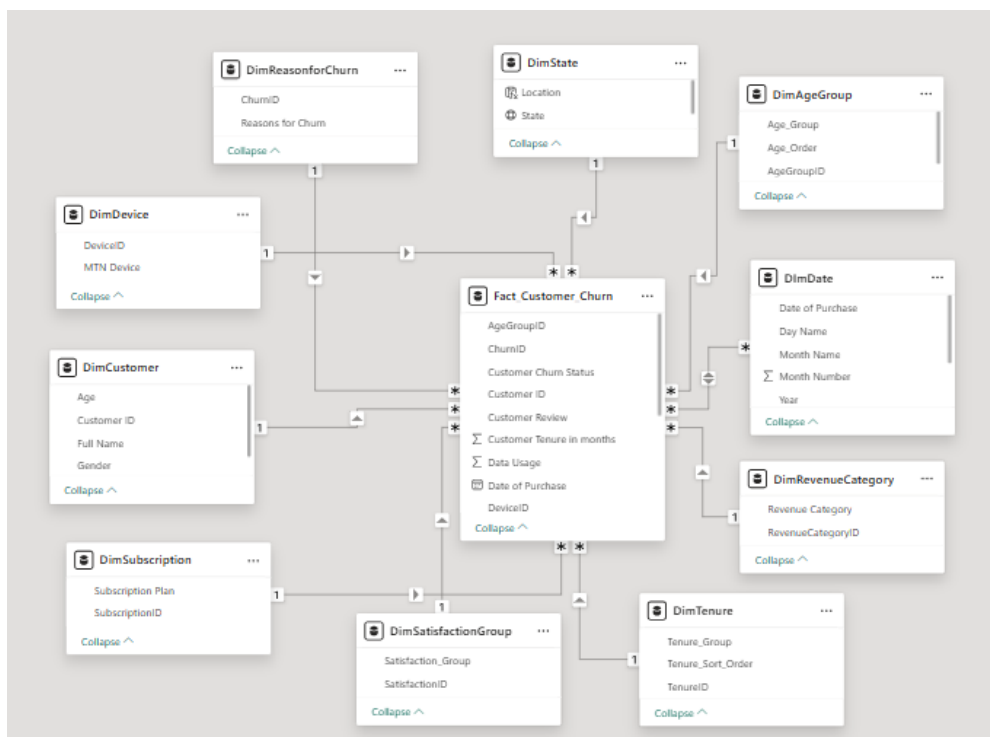
Relationships between the fact table and dimension tables were implemented using **one-to-many relationships**, with the dimension tables positioned on the "one" side and the fact table on the "many" sides. A single-direction filter propagation was adopted to maintain data integrity and improve model performance.

The implementation of the Star Schema provides several important advantages, including:

- a) Improved report performance through efficient relationship management.
- b) Simplified DAX measure development and maintenance.
- c) Reduced data redundancy and improved scalability.
- d) Enhanced filtering and slicing across multiple dashboards.
- e) Support for advanced Business Intelligence calculations and time intelligence functions.
- f) Improved flexibility for future expansion of the analytical model.

The resulting data model provides a robust analytical framework that enables interactive dashboard reporting while supporting accurate and consistent business analysis across all five dashboards developed in this project.

Figure 2.3: Star Schema Data Model for the MTN Customer Churn Analysis Dashboard



CHAPTER THREE

EXECUTIVE DASHBOARD ANALYSIS

The Executive Overview Dashboard provides a consolidated view of MTN's overall operational, customer, and financial performance. Designed as a strategic decision-support tool, the dashboard integrates key performance indicators with interactive visualizations to enable management to monitor business health, evaluate customer behavior, identify churn patterns, and assess revenue performance. Rather than relying on multiple operational reports, executives can use this dashboard to obtain a real-time overview of business performance while identifying areas requiring immediate managerial attention.

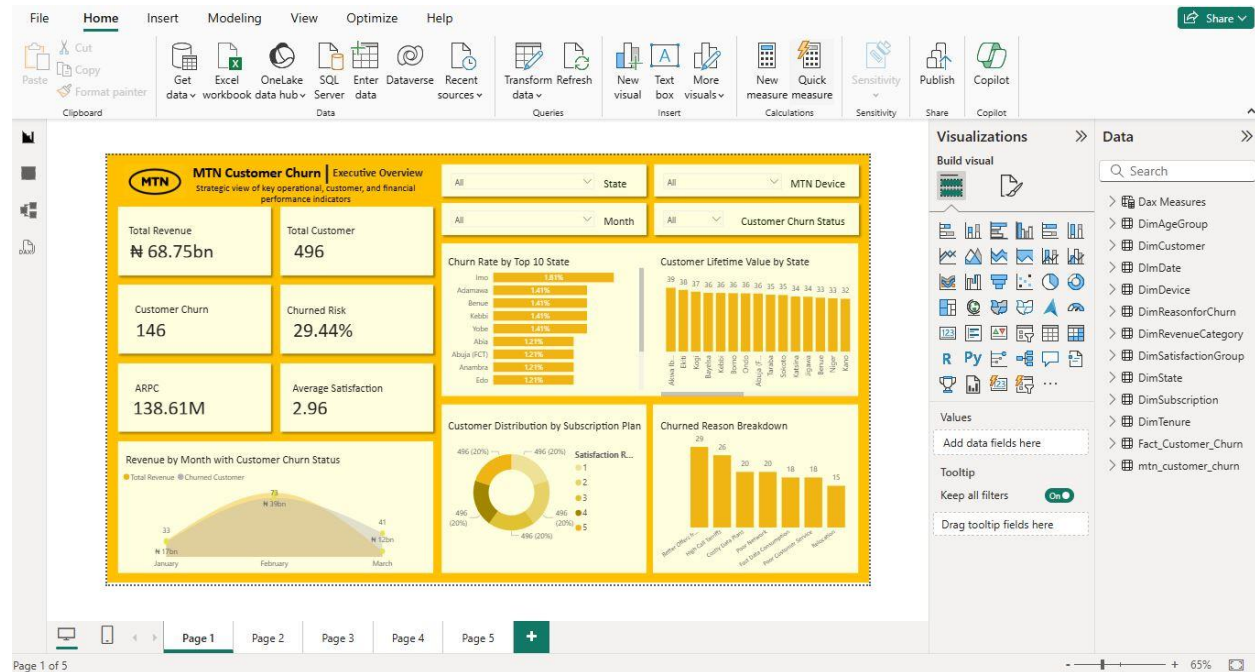


Figure 3.1: Executive Overview Dashboard

The dashboard reveals that MTN generated a total revenue of 68.75 billion from 496 customers during the reporting period. However, despite the strong financial performance, 146 customers were identified as churned, representing a 29.44% churn rate, which poses a significant threat to long-term revenue sustainability. Furthermore, the average customer satisfaction score of 2.96 out of 5 suggests that service quality and customer experience require improvement. Collectively, these indicators demonstrate that although the organization maintains strong revenue generation, customer retention remains a critical business challenge requiring strategic intervention.

3.1 Customer Lifetime Value by State

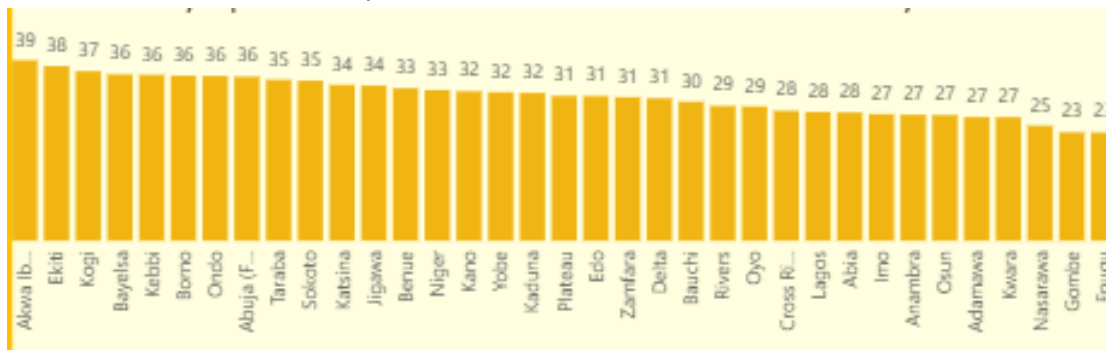


Figure 3.2: Customer Lifetime Value by State

Business Insight

Customer Lifetime Value (CLV) analysis reveals significant differences in long-term customer profitability across MTN's regional markets. The visualization demonstrates that certain states consistently generate substantially higher lifetime value than others, indicating that customer profitability is influenced by regional purchasing behavior, economic conditions, product adoption, and customer engagement rather than customer volume alone.

From an executive perspective, this finding challenges the traditional assumption that acquiring more customers automatically results in greater profitability. Instead, the dashboard demonstrates that the financial value generated by each customer varies considerably across geographical markets.

Consequently, strategic success should be measured by customer value creation rather than customer acquisition alone.

Root Cause Analysis

Several business factors explain these regional differences. States exhibiting higher Customer Lifetime Value are likely characterized by stronger purchasing power, greater adoption of premium subscription plans, higher digital service utilization, and longer customer retention. In contrast, lower-performing states may experience higher customer churn, lower product adoption, weaker network coverage, or increased competitive pressure from rival telecommunications providers. These findings indicate that geographical performance is driven by a combination of customer behavior and operational effectiveness.

Business Risk

Applying a uniform commercial strategy across all states may result in inefficient resource allocation and lower returns on investment. High-value regions may become vulnerable to competitive acquisition if they do not receive sufficient investment, while continued spending in low-value regions without targeted improvement initiatives may reduce overall profitability. Without regional performance intelligence, MTN risks allocating marketing budgets, network investments, and customer retention programs without maximizing financial return.

Strategic Impact

Protecting customers in high-value states delivers greater long-term financial benefits than focusing exclusively on expanding the customer base. Increasing retention within these regions directly improves Customer Lifetime Value, recurring revenue, Average Revenue per User (ARPU), and long-term profitability while reducing customer acquisition costs. Consequently, regional customer value should become a key performance indicator within MTN's commercial planning framework.

Customer Churn by State

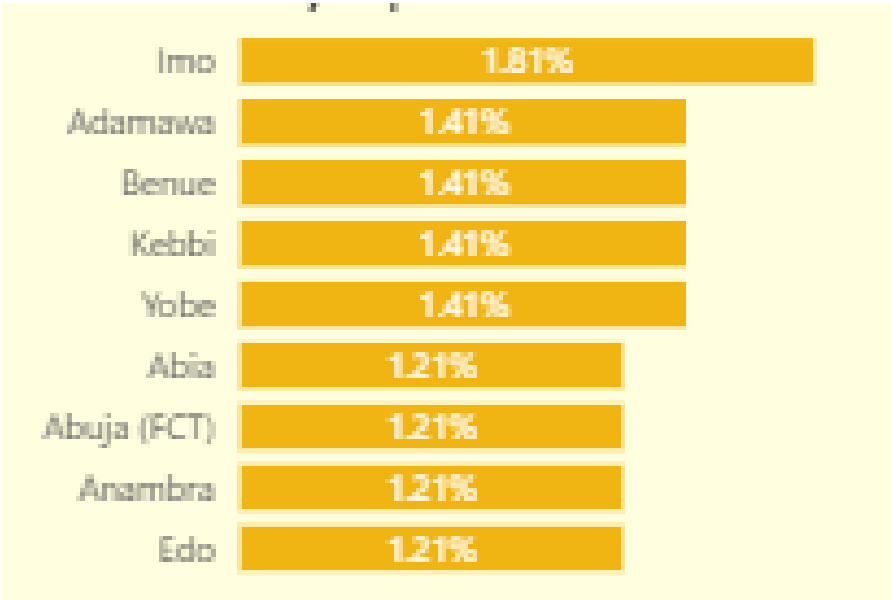


Figure 3.3: Customer Churn by State

Business Insight

The geographical distribution of customer churn reveals that customer attrition is not evenly dispersed across MTN's operational markets. Instead, the dashboard demonstrates that a relatively small number of states account for a disproportionately high percentage of customer losses. This finding indicates that customer churn is driven by localized operational and commercial challenges rather than representing a nationwide decline in customer satisfaction.

From an executive perspective, this insight fundamentally changes how customer retention should be managed. Rather than implementing expensive nationwide retention campaigns, MTN has an opportunity to focus resources on high-risk regions where targeted interventions are likely to produce significantly greater returns. The dashboard therefore transforms customer churn from a national performance indicator into a regional strategic management issue.

Root Cause Analysis

Several factors may explain why customer churn is concentrated within specific states. Regional differences in network quality, service availability, competitive intensity, customer purchasing power, and responsiveness to customer complaints can all influence customer retention outcomes. Markets

experiencing weaker infrastructure or stronger competitor presence are naturally more vulnerable to customer migration.

The concentration of churn within selected states also suggests that operational performance is inconsistent across MTN's geographical footprint. This inconsistency presents an opportunity for management to investigate state-level operational practices and replicate successful approaches across lower-performing regions.

Business Risk

Persistent customer losses within key geographical markets present both financial and strategic risks. High churn reduces recurring revenue, increases customer acquisition costs, and weakens MTN's competitive position within affected regions. If left unresolved, competitors may strengthen their market presence by acquiring dissatisfied customers, making future recovery significantly more expensive. Moreover, continued underperformance within strategic states could negatively affect national revenue growth and reduce the return on investments made in network infrastructure and customer acquisition.

Strategic Impact

Reducing churn within the highest-risk states would deliver a disproportionate financial benefit compared with broad national retention programs. Every percentage reduction in customer attrition preserves recurring subscription revenue, increases Customer Lifetime Value (CLV), and reduces marketing expenditure associated with replacing lost customers.

Consequently, regional churn management should become a strategic performance indicator monitored by executive leadership on a continuous basis.

Customer Satisfaction Score



Figure 3.4: Customer Satisfaction KPI

Business Insight

The Customer Satisfaction Score represents one of the most influential leading indicators of MTN's future business performance. While financial metrics measure historical success, customer satisfaction predicts future customer behavior, including loyalty, repeat purchases, and customer churn. The dashboard reports an average customer satisfaction score of 2.96, indicating that customer experience falls below the level expected for sustained competitive advantage.

From an executive perspective, this finding extends beyond customer service performance. Low

customer satisfaction directly influences revenue stability because dissatisfied customers are significantly more likely to reduce spending, switch to competing providers, and discourage prospective customers through negative word-of-mouth. Consequently, customer satisfaction should be recognized as a strategic business performance indicator rather than solely an operational service metric.

Root Cause Analysis

The satisfaction score reflects the cumulative effect of several customer experience factors, including network reliability, pricing competitiveness, complaint resolution efficiency, subscription value, and overall service quality. Earlier dashboard findings indicate that many customers attribute churn to poor customer service, expensive data plans, and better offers from competitors. These issues collectively contribute to declining customer satisfaction. The analysis therefore suggests that customer dissatisfaction is systemic rather than isolated, requiring coordinated improvements across multiple business functions.

Business Risk

Failure to improve customer satisfaction will likely accelerate customer churn, reduce customer loyalty, and weaken MTN's market position. Dissatisfied customers are less likely to purchase premium products, recommend MTN to others, or remain loyal during periods of competitive pricing pressure. Over time, declining satisfaction may also increase customer acquisition costs as the organization invests more resources replacing customers who could otherwise have been retained.

Strategic Impact

Customer satisfaction has a direct relationship with revenue growth, Customer Lifetime Value, Average Revenue per User (ARPU), and profitability. Even modest improvements in satisfaction can generate substantial long-term financial benefits by increasing customer retention, encouraging premium product adoption, and strengthening brand loyalty.

Improving customer satisfaction should therefore be viewed as a revenue growth initiative rather than simply a customer service objective.

Revenue Performance (Total Revenue KPI)



Figure 3.5: Total Revenue KPI

Business Insight

Total Revenue represents MTN's overall financial performance during the reporting period and serves as the primary indicator of the organization's commercial success. While the reported revenue demonstrates that the business continues to generate substantial income from its customer base, the true executive value of this metric lies in understanding the drivers that sustain or threaten future revenue growth.

Revenue should not be viewed as an isolated financial outcome but as the cumulative result of customer acquisition, customer retention, pricing strategy, subscription adoption, service quality, and customer satisfaction. The Executive Dashboard indicates that MTN has established a strong revenue foundation; however, earlier analyses reveal emerging risks associated with customer churn and below-average customer satisfaction. These findings suggest that current revenue performance may not be sustainable unless proactive measures are implemented.

Root Cause Analysis

Revenue performance has been influenced by several interconnected business drivers. The adoption of premium subscription plans, increasing demand for broadband connectivity, and consistent customer purchases have all contributed positively to financial performance. However, these gains are partially offset by customer attrition, competitive pricing pressures, and service-related dissatisfaction, which reduce recurring revenue opportunities. The analysis also indicates that revenue growth is concentrated within specific customer segments and product categories. Such concentration suggests that the organization has not yet fully diversified its revenue sources.

Business Risk

Strong current revenue can create a false sense of security if management focuses exclusively on financial outcomes rather than the operational factors driving those outcomes. Without improving customer retention and satisfaction, MTN may experience declining recurring revenue, increased customer acquisition costs, and reduced profitability over time. In addition, overreliance on a limited number of premium products increases exposure to competitive actions and changing customer preferences.

Strategic Impact

Revenue stability directly influences investment capacity, shareholder value, infrastructure expansion, and long-term business sustainability. Protecting existing revenue streams through improved customer retention is significantly more cost-effective than replacing lost customers through new acquisition campaigns. Furthermore, increasing Customer Lifetime Value (CLV) and Average Revenue per User (ARPU) offers a more sustainable path to profitability than relying solely on subscriber growth.

Customer Base Performance (Total Customers KPI)



Figure 3.6: Total Customers KPI

Business Insight

The Total Customers KPI reflects the size of MTN's active customer portfolio and provides an essential measure of market reach. However, from an executive perspective, the strategic importance of this metric extends beyond customer volume. The true competitive advantage lies not in acquiring the largest customer base but in maximizing the economic value generated by each customer. The Executive Dashboard demonstrates that MTN has established a substantial customer portfolio capable of generating significant revenue. Nevertheless, previous analyses indicate that customer profitability varies considerably across demographic groups, geographical regions, and product categories. This finding reinforces the principle that customer quality is as important as customer quantity. Executives should therefore interpret customer growth in conjunction with Customer Lifetime Value, retention performance, and revenue contribution rather than viewing subscriber numbers as an independent measure of success.

Root Cause Analysis

Customer growth has been supported by MTN's extensive market presence, diverse subscription offerings, and established brand recognition. However, differences in customer engagement, purchasing behavior, and service utilization mean that not all subscribers contribute equally to organizational profitability.

Furthermore, customer churn continues to reduce the long-term value of newly acquired subscribers, emphasizing the need for stronger retention initiatives.

Business Risk

Prioritizing customer acquisition without corresponding improvements in customer retention may increase marketing expenditure while limiting long-term profitability. High customer turnover erodes the financial benefits of acquisition campaigns and weakens customer lifetime value.

Failure to improve the value generated from existing customers may also constrain future revenue growth despite continued expansion of the subscriber base.

Strategic Impact

Increasing the value of existing customers delivers greater financial returns than relying exclusively on new customer acquisition. Improvements in customer engagement, product adoption, and retention strengthen recurring revenue while reducing acquisition costs.

Consequently, management should evaluate customer performance using value-based metrics rather than subscriber counts alone.

Customer Churn Performance (Churn Rate KPI)



Figure 3.7: Customer Churn KPI

Business Insight

Customer churn is one of the most significant strategic indicators within the telecommunications industry because it directly measures the organization's ability to retain profitable customer relationships. Unlike revenue, which reflects historical performance, churn provides an early indication of future commercial stability.

The Executive Dashboard indicates that customer attrition remains a critical challenge requiring immediate executive attention. Every customer lost represents not only an immediate reduction in recurring revenue but also the loss of future purchases, reduced customer lifetime value, and additional acquisition costs required to replace that customer. From a strategic perspective, the dashboard demonstrates that customer retention should be viewed as a primary driver of sustainable business growth rather than solely an operational responsibility.

Root Cause Analysis

Customer churn is influenced by a combination of service quality, pricing competitiveness, customer experience, network reliability, and competitor activity. The dashboard indicates that dissatisfaction with customer service and attractive competitor offerings are significant contributors to customer migration. These findings suggest that churn is driven by both internal operational challenges and external market competition.

Business Risk

Persistent customer churn reduces recurring revenue, increases marketing expenditure, and weakens

competitive positioning. As customer acquisition costs continue to rise across the telecommunications industry, replacing lost customers becomes progressively more expensive than retaining existing ones. Failure to reduce churn may therefore limit revenue growth despite successful customer acquisition campaigns.

Strategic Impact

Even modest reductions in churn can produce substantial financial benefits through increased recurring revenue, higher Customer Lifetime Value, improved Average Revenue per User, and lower acquisition costs. Consequently, customer retention initiatives often deliver one of the highest returns on investment within the telecommunications sector.

CHAPTER FOUR

Customer Demography Analysis Dashboard

Customer behavior has become one of the most valuable strategic assets within the telecommunications industry. While financial performance reflects the outcome of business activities, customer behavior explains the underlying factors that determine long-term organizational success. Understanding who customers are, how they interact with products and services, and what influences their purchasing and retention decisions enables organizations to design more effective commercial strategies and strengthen competitive advantage.

The Customer Analysis Dashboard was developed to provide MTN's executive management with a comprehensive understanding of customer demographics, behavioral patterns, and engagement characteristics. Rather than presenting demographic statistics alone, the dashboard transforms customer information into strategic business intelligence that supports customer segmentation, personalized marketing, product development, and customer retention planning. This chapter interprets customer behavior from an executive perspective by examining the strategic insights generated through customer demographics, tenure distribution, age segmentation, device adoption, and geographical patterns. The objective is not merely to describe customer characteristics but to identify opportunities for revenue growth, improve customer lifetime value, reduce churn risk, and strengthen MTN's long-term market position.

Customer Analysis Dashboard

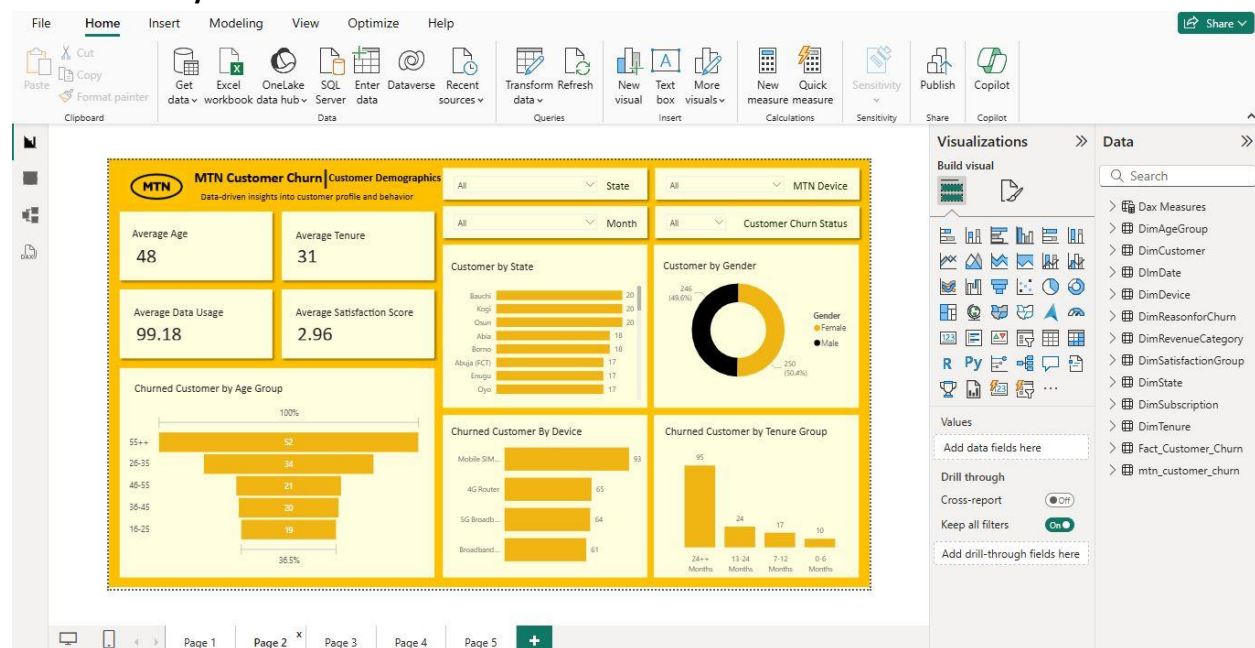


Figure 4.1: Customer Analysis Dashboard

The dashboard integrates demographic indicators, customer behavioral metrics, and product usage patterns into a single analytical environment. It enables executive management to evaluate customer composition, purchasing behavior, engagement levels, and risk characteristics that influence long-term revenue generation and customer retention.

Gender Distribution

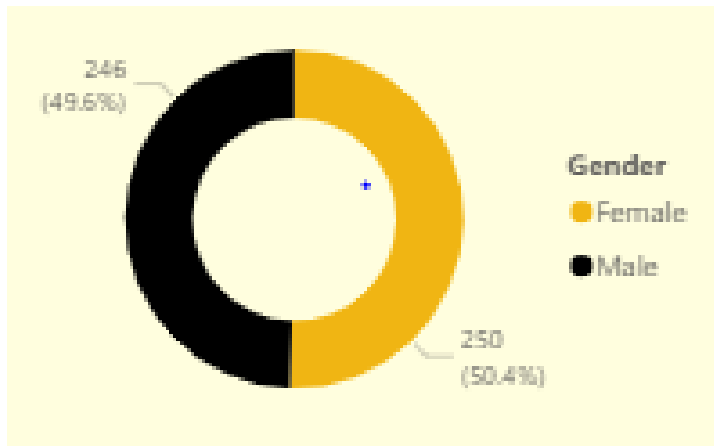


Figure 4.2: Customer Distribution by Gender

Business Insight

The gender distribution analysis indicates that MTN serves a relatively balanced customer population, with male and female subscribers contributing almost equally to the overall customer base. From an executive perspective, this balance demonstrates that the company's products and services possess broad market appeal rather than depending on a single demographic segment for commercial success. However, the strategic value of this visualization extends beyond representation. Equal gender participation suggests that future revenue growth is unlikely to be achieved by targeting one gender over another. Instead, sustainable growth will depend on understanding behavioral differences, purchasing habits, product preferences, and digital consumption patterns within each demographic group. The dashboard therefore shifts management's attention away from demographic equality toward behavioral profitability.

Root Cause Analysis

The balanced distribution reflects MTN's nationwide market presence and diversified telecommunications portfolio. Products such as voice services, broadband internet, digital payment solutions, and enterprise connectivity appeal broadly across both male and female customers. The absence of significant gender imbalance also indicates that barriers to product accessibility are relatively low, allowing the organization to compete effectively across diverse customer segments.

Business Risk

Although gender representation is balanced, treating both groups as homogeneous customer segments may reduce marketing effectiveness. Customer preferences, purchasing priorities, and digital engagement behaviors often differ across demographic groups. Failure to understand these behavioral differences may result in generic marketing campaigns that fail to maximize customer engagement and revenue potential.

Strategic Impact

Behavioral segmentation based on purchasing patterns rather than demographic characteristics enables more effective allocation of marketing resources, improves campaign conversion rates, and increases Customer Lifetime Value (CLV). Personalized customer engagement also strengthens loyalty while reducing customer acquisition costs.

Customer Age Group Analysis

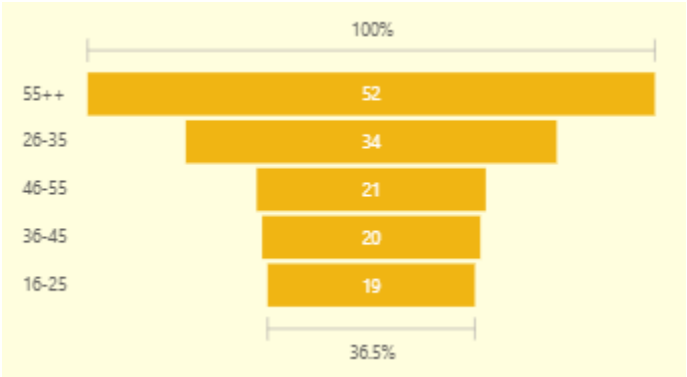


Figure 4.4: Customer Distribution by Age Group

Business Insight

The age group analysis provides strategic insight into the demographic composition of MTN's customer base and highlights how different customer segments contribute to long-term business sustainability. While the dashboard indicates that customers are distributed across multiple age categories, the concentration of subscribers within specific age groups demonstrates that revenue generation and customer engagement are strongly influenced by demographic characteristics.

From an executive perspective, age segmentation should not be interpreted merely as a demographic statistic but as a predictor of purchasing behavior, digital adoption, product preference, and customer lifetime value. Different age groups exhibit distinct consumption patterns, communication needs, and responsiveness to marketing initiatives. Consequently, understanding these behavioral differences enables MTN to design customer-centric products that improve profitability while strengthening long-term customer relationships. The analysis suggests that middle-aged customers represent the organization's most commercially valuable segment due to their relatively stable purchasing power and consistent utilization of telecommunications services. However, younger customers represent future growth opportunities because of their increasing dependence on digital connectivity, online entertainment, and mobile financial services.

Root Cause Analysis

Customer behavior changes significantly throughout different stages of life. Younger subscribers typically demonstrate higher demand for mobile data, digital entertainment, gaming, and social networking services. Working professionals require reliable connectivity for business operations, remote work, online meetings, and financial transactions. Older customers generally prioritize network reliability, customer support, and service consistency over high-volume data consumption. These behavioral differences naturally influence purchasing decisions, subscription plan selection, and overall customer

value.

Business Risk

Failure to recognize demographic differences may result in generic commercial strategies that fail to meet the needs of diverse customer segments. Marketing campaigns designed for the entire customer population are often less effective than personalized approaches tailored to specific demographic groups.

In addition, an overdependence on one age segment exposes MTN to long-term market risks if changing consumer preferences or demographic shifts reduce demand within that group.

Strategic Impact

Age-based customer segmentation enables MTN to optimize product design, pricing strategies, and promotional campaigns. Delivering age-specific offerings increases customer satisfaction, strengthens brand loyalty, improves Average Revenue per User (ARPU), and enhances Customer Lifetime Value (CLV). Furthermore, retaining younger customers at an early stage of their relationship with MTN increases the probability of generating recurring revenue over many years.

Customer Tenure Analysis

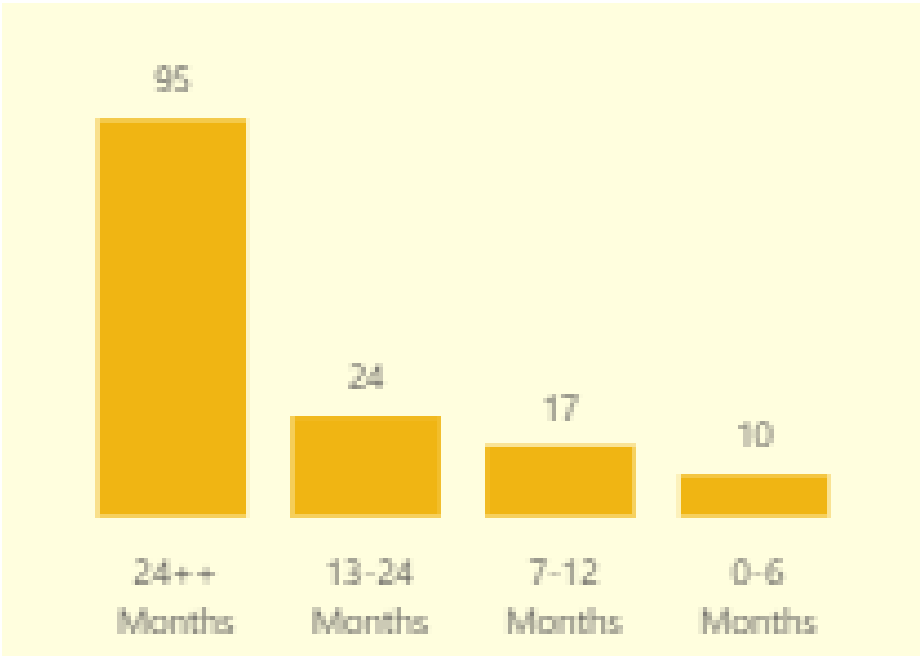


Figure 4.5: Customer Tenure Distribution

Business Insight

Customer tenure is one of the strongest indicators of customer loyalty and long-term profitability. The dashboard demonstrates how long subscribers have remained with MTN, providing management with valuable insight into the organization's ability to establish durable customer relationships. From an executive perspective, tenure is directly linked to Customer Lifetime Value. Customers who remain with MTN for longer periods generally contribute more revenue through repeated purchases, increased

product adoption, and greater engagement with digital services. Consequently, customer tenure should be viewed as a strategic asset rather than simply a descriptive customer characteristic.

The dashboard indicates that MTN maintains a substantial proportion of long-tenured customers, reflecting strong brand recognition and customer loyalty. Nevertheless, the presence of newer customers highlights the importance of maintaining effective onboarding and early engagement strategies.

Root Cause Analysis

Long customer tenure is typically associated with positive service experiences, reliable network performance, competitive pricing, and strong customer relationships. Conversely, customers with shorter tenure often remain within a critical evaluation period during which they compare MTN's services with competing providers.

The analysis therefore demonstrates that customer experience during the first months of subscription significantly influences long-term retention outcomes.

Business Risk

The first twelve months of the customer lifecycle represent the period of greatest retention risk. Failure to engage customers effectively during this stage increases the likelihood of early churn, reducing Customer Lifetime Value and increasing acquisition costs.

Additionally, declining tenure trends may indicate underlying service quality issues that require immediate executive attention.

Strategic Impact

Increasing average customer tenure generates significant financial benefits by strengthening recurring revenue, reducing marketing expenditure, and improving overall profitability. Long-tenured customers also exhibit higher adoption of premium services, increasing Average Revenue per User (ARPU) and Customer Lifetime Value (CLV).

Consequently, customer tenure represents one of the organization's most valuable long-term financial assets.

Device Preference Analysis

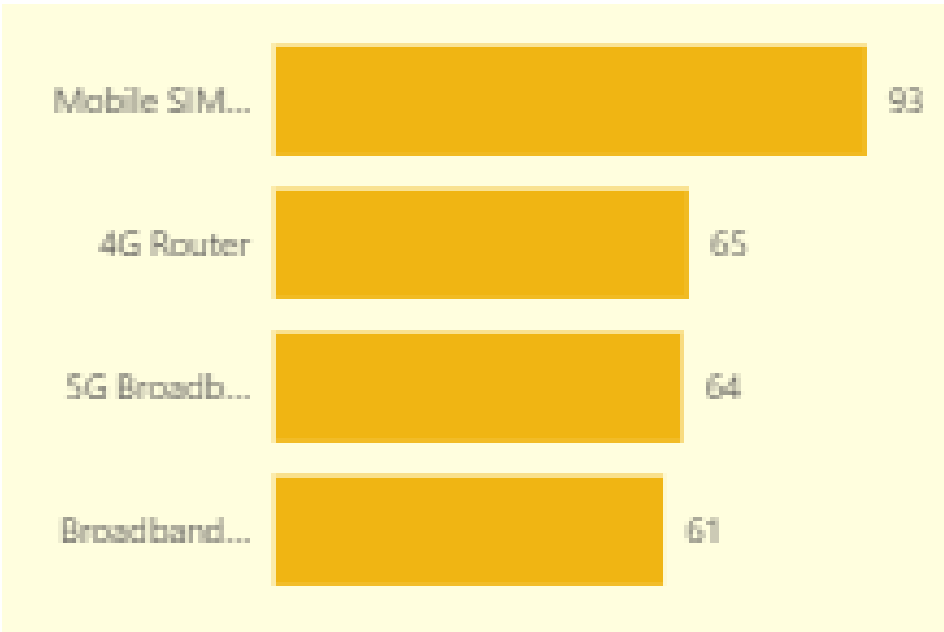


Figure 4.6: Customer Distribution by Device Type

Business Insight

Device preference analysis provides strategic insight into how customers access MTN's telecommunications services and reveals changing patterns in digital consumption. The dashboard demonstrates that customer demand varies considerably across Mobile SIM Cards, Broadband MiFi devices, 4G Routers, and 5G Broadband Routers, reflecting evolving connectivity requirements within Nigeria's digital economy. The increasing adoption of broadband-enabled devices indicates that customer expectations are shifting from traditional mobile communication toward high-speed internet connectivity capable of supporting business operations, remote learning, entertainment, and digital innovation.

For executive management, this transition represents both a commercial opportunity and a strategic transformation of MTN's business model.

Root Cause Analysis

Rapid digital transformation has significantly increased customer demand for reliable broadband infrastructure. Growth in video streaming, cloud services, online education, digital banking, and remote work has encouraged migration toward devices capable of delivering higher network performance. The continued presence of Mobile SIM users reflects affordability considerations and widespread smartphone adoption, particularly among price-sensitive customer segments.

Business Risk

Failure to expand broadband infrastructure in line with customer demand could reduce MTN's competitiveness and encourage migration to alternative providers offering superior connectivity

solutions. Likewise, excessive dependence on traditional Mobile SIM services may limit future revenue growth as customer expectations continue evolving.

Strategic Impact

Broadband devices typically generate higher Average Revenue per User (ARPU) and stronger Customer Lifetime Value than traditional mobile services. Increasing broadband adoption therefore provides an effective strategy for expanding revenue while maximizing the value generated from existing customers.

Customer Geographical Distribution

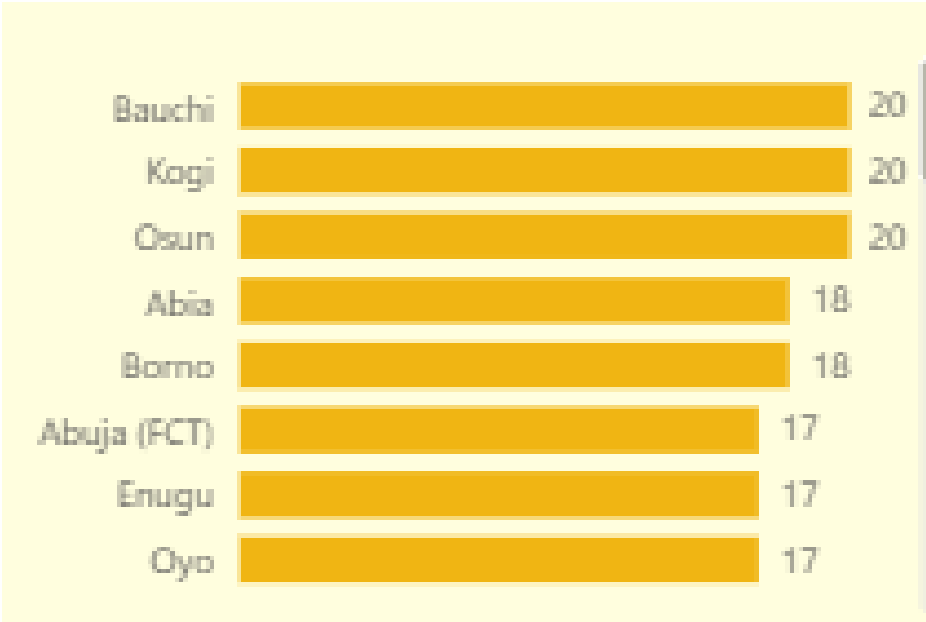


Figure 4.7: Customer Distribution by State

Business Insight

The geographical distribution of MTN's customer base demonstrates that customer acquisition and market penetration vary significantly across different states. While a broad national presence strengthens MTN's competitive position, the uneven distribution of customers highlights differences in regional market opportunities, economic activities, infrastructure development, and digital adoption. From an executive perspective, this visualization is more than a geographical representation of customers; it is a strategic indicator of market concentration and future expansion opportunities. States with larger customer populations represent established revenue centers where protecting market share is critical. Conversely, states with relatively lower customer penetration may represent untapped markets capable of supporting future business growth through targeted commercial investments. The analysis demonstrates that customer distribution should directly influence strategic planning, network expansion, marketing investment, and regional sales objectives.

Root Cause Analysis

Several factors contribute to the observed geographical differences. Urbanized states typically benefit

from higher population density, stronger economic activity, improved telecommunications infrastructure, and greater adoption of digital technologies. These conditions naturally encourage higher customer acquisition and greater demand for broadband services. In contrast, states with lower customer representation may experience infrastructure limitations, lower digital awareness, reduced purchasing power, or stronger competition from alternative network providers.

These regional variations demonstrate that customer acquisition performance is influenced by external market conditions as much as internal commercial strategies.

Business Risk

Applying identical marketing and investment strategies across all states may result in inefficient resource allocation. Regions with strong growth potential may receive insufficient investment, while excessive expenditure in saturated or lower-performing markets may generate diminishing returns. Failure to recognize regional market dynamics could also reduce MTN's ability to respond effectively to competitive threats and changing customer expectations within specific locations.

Strategic Impact

A geographically optimized commercial strategy enables MTN to maximize return on investment by directing marketing expenditure, network development, and customer engagement initiatives toward regions capable of delivering the highest commercial value.

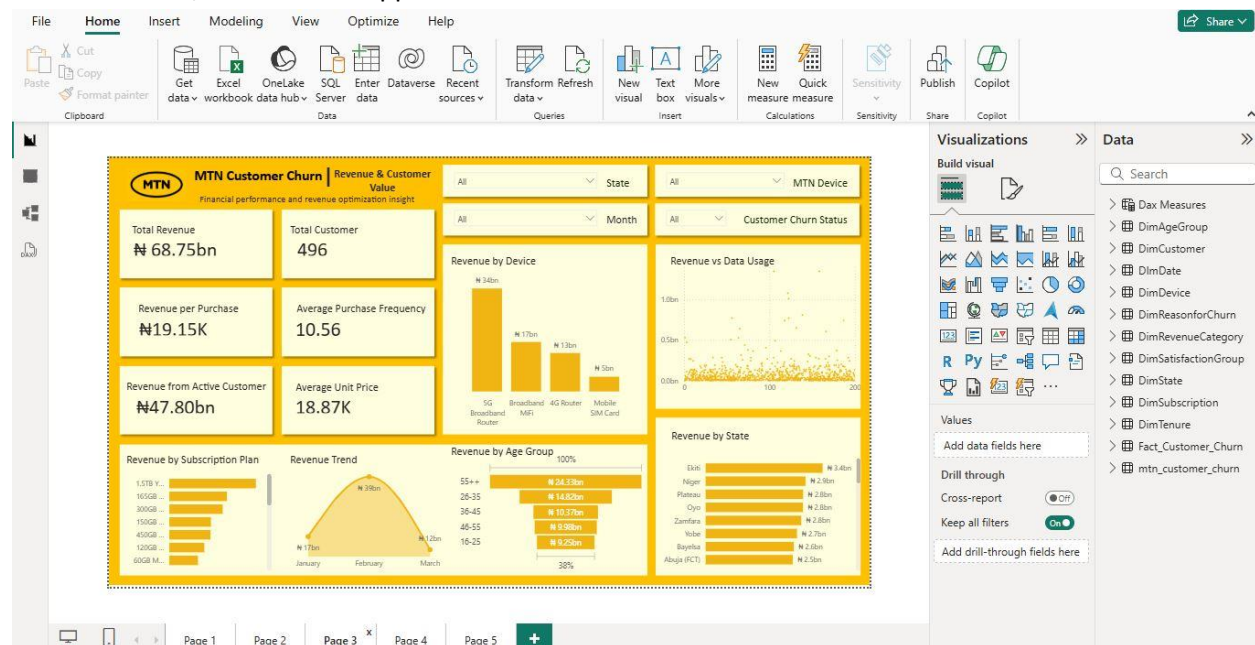
Strengthening market penetration within high-potential states increases customer acquisition, improves regional revenue performance, and enhances long-term profitability while reducing inefficient operational spending.

CHAPTER FIVE

Revenue and Customer Value Dashboard Analysis

Revenue is the most visible indicator of organizational performance; however, sustainable financial success depends not only on the amount of revenue generated but also on understanding the underlying factors that drive revenue growth. Within the telecommunications industry, organizations must continuously evaluate customer purchasing behavior, product profitability, pricing effectiveness, and customer lifetime value to remain competitive in an increasingly dynamic market. The Revenue and Customer Value Dashboard was developed as an executive decision-support tool to enable MTN management to evaluate the financial performance of its products, customers, and geographical markets. Rather than reporting revenue as a single financial figure, the dashboard integrates multiple dimensions of business performance, including revenue contribution by product, customer purchasing behavior, geographical revenue distribution, demographic spending patterns, and customer value indicators.

The objective of this dashboard is to transform transactional data into actionable business intelligence capable of supporting strategic planning, commercial decision-making, and long-term profitability. Through interactive visualizations and financial performance indicators, executives can identify the products generating the highest returns, recognize profitable customer segments, monitor changing revenue trends, and discover opportunities to increase customer lifetime value.



Consequently, this chapter presents a comprehensive business analysis of MTN's revenue performance by examining the strategic insights generated from each visualization. Rather than describing dashboard components, the analysis focuses on interpreting business performance, identifying commercial risks, evaluating financial implications, and proposing evidence-based recommendations capable of strengthening organizational profitability and sustaining competitive advantage.

Revenue by Device

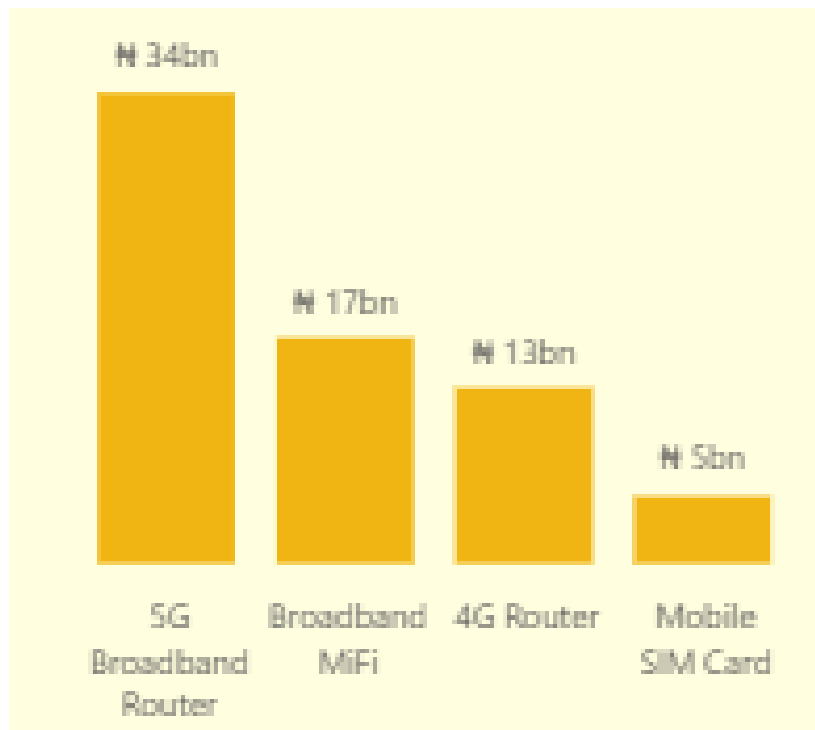


Figure 5.2: Revenue by Device

Business Insight

The Revenue by Device analysis demonstrates that MTN's revenue generation is heavily concentrated within its premium broadband product portfolio. The dashboard reveals that the 5G Broadband Router contributes the largest proportion of total device revenue, significantly outperforming Broadband MiFi, 4G Router, and Mobile SIM services. This indicates that customer spending is increasingly shifting toward high-capacity internet solutions capable of supporting data-intensive activities such as remote work, digital entertainment, online education, cloud computing, and enterprise connectivity.

This finding reflects a structural transformation within Nigeria's telecommunications market, where broadband connectivity is becoming a primary source of commercial value rather than traditional voice-based communication. Customers are demonstrating a willingness to invest in premium internet services that provide superior speed, reliability, and user experience. Consequently, broadband products have evolved from complementary services into strategic revenue drivers capable of sustaining long-term business growth.

Root Cause Analysis

Several commercial and technological factors contribute to this revenue concentration. The rapid growth of digital lifestyles has significantly increased household and business demand for reliable broadband connectivity. Simultaneously, the expansion of remote working arrangements, online learning platforms, and streaming services has accelerated customer migration toward high-capacity internet products. Conversely, Mobile SIM services continue to generate comparatively lower revenue despite serving a larger customer population. This disparity suggests that many subscribers remain within lower-value

pricing tiers and have not transitioned into premium digital products. While customer acquisition remains strong, revenue optimization within the traditional subscriber base appears limited.

Business Risk

Although the outstanding performance of broadband products represents a commercial success, excessive dependence on a limited number of premium offerings introduces strategic risk. Any deterioration in network performance, aggressive competitive pricing, regulatory intervention, or technological disruption affecting broadband services could significantly reduce organizational revenue. Furthermore, underperformance within Mobile SIM products indicates unrealized revenue potential. Failure to increase customer value within this segment may constrain future profitability as market competition intensifies.

Financial Impact

The dashboard indicates that broadband devices generate substantially higher financial returns per customer than conventional Mobile SIM services. This suggests that increasing broadband adoption could improve Average Revenue per Customer (ARPC), strengthen Customer Lifetime Value (CLV), and increase recurring revenue without proportionally increasing customer acquisition costs. Therefore, future revenue growth should be driven by customer value enhancement rather than subscriber growth alone.

CHAPTER SIX

Churn Deep Dive: Root Cause Analysis of Customer Attrition

Customer churn remains one of the most critical challenges facing the telecommunications industry because every lost customer represents a reduction in recurring revenue, customer lifetime value, and long-term profitability. In highly competitive markets such as Nigeria's telecommunications sector, retaining existing customers is significantly more cost-effective than acquiring new ones. Consequently, understanding why customers leave has become a strategic priority for executive management. While previous chapters examined customer demographics, revenue performance, and overall business indicators, this chapter focuses specifically on identifying the underlying drivers of customer attrition. The Churn Deep Dive Dashboard transforms customer data into actionable intelligence by analyzing churn behavior across subscription plans, customer satisfaction levels, geographical regions, and financial impact.

The objective of this chapter is to identify the operational, commercial, and customer experience factors responsible for churn, quantify their financial implications, and recommend evidence-based strategies capable of improving customer retention and protecting long-term business value.

Churn Deep Dive Dashboard

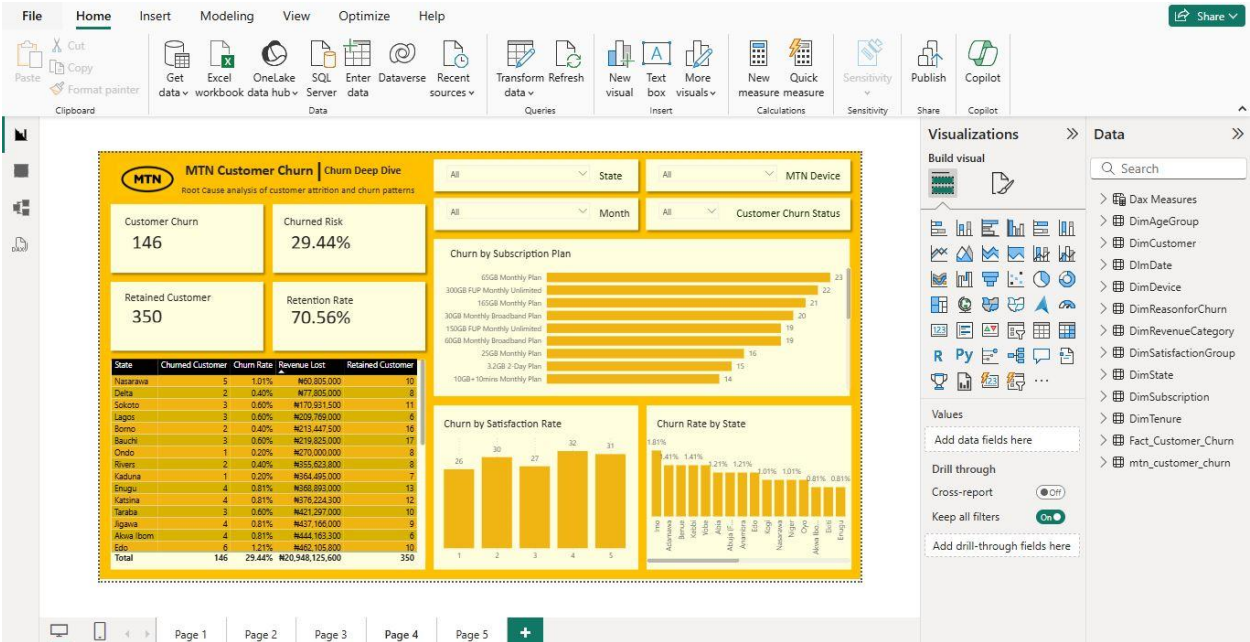


Figure 6.1: Churn Deep Dive Dashboard

The Churn Deep Dive Dashboard consolidates customer attrition metrics, revenue loss, retention indicators, and behavioral patterns into a single analytical environment. Unlike traditional reporting dashboards, it enables executive management to investigate the root causes of churn, quantify associated financial losses, and prioritize strategic interventions based on business impact.

Customer Churn KPI

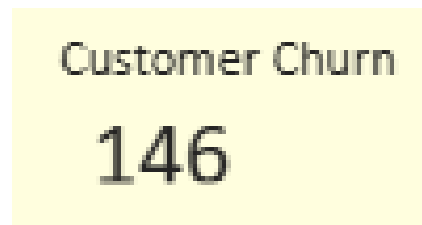


Figure 6.2: Customer Churn KPI

Business Insight

The reports show that out of 496 customers 146 have churned, representing a 29.44% churn rate. This indicates that nearly three out of every ten customers* are no longer retained by MTN during the reporting period. From an executive perspective, this is not simply an operational metric—it is a direct indicator of future revenue erosion. Every churned customer represents the loss of recurring subscription income, reduced opportunities for cross-selling premium products, and additional acquisition costs required to replace lost subscribers.

The churn rate observed in this analysis signals that customer retention requires immediate executive attention. Although MTN continues to maintain a strong customer base, sustaining future profitability will depend on reducing preventable customer losses.

Root Cause Analysis

The dashboard suggests that churn is influenced by multiple interconnected factors, including pricing concerns, customer dissatisfaction, competitor offerings, and perceived service quality. Rather than being caused by a single issue, customer attrition reflects weaknesses across the customer experience lifecycle.

Business Risk

Persistent churn at this level threatens recurring revenue, increases customer acquisition expenditure, and weakens long-term market competitiveness. If left unaddressed, customer losses may offset future gains achieved through acquisition campaigns.

Customer Retention Rate



Figure 6.3: Retention Rate KPI

Business Insight

The dashboard reports a 70.56% customer retention rate, indicating that MTN successfully retains approximately seven out of every ten customers. Although this demonstrates a relatively stable customer base, the remaining 29.44% attrition highlights a significant opportunity for improvement. Retention should not be viewed merely as preserving customer numbers but as protecting recurring revenue, customer loyalty, and long-term profitability. High-performing telecommunications companies continuously improve retention because retained customers typically generate greater lifetime value than newly acquired subscribers.

Business Implication

Improving retention by only a few percentage points could substantially increase annual revenue while reducing acquisition costs.

Churn by Subscription Plan



Figure 6.4: Churn by Subscription Plan

Business Insight

The analysis reveals that customer churn is concentrated within selected subscription plans rather than being evenly distributed across the product portfolio. This finding suggests that certain products fail to meet customer expectations regarding affordability, perceived value, or service quality. High churn within specific plans indicates that customers are evaluating competing offers and migrating toward alternatives that better satisfy their communication and digital connectivity needs.

Root Cause Analysis

Several factors may explain product-specific churn, including pricing structures, limited data allocations, network experience, and insufficient differentiation from competitor offerings.

Business Risk

Products experiencing consistently high churn reduce recurring revenue and weaken customer loyalty. Maintaining underperforming plans without redesign may accelerate customer migration.

Churn by Customer Satisfaction

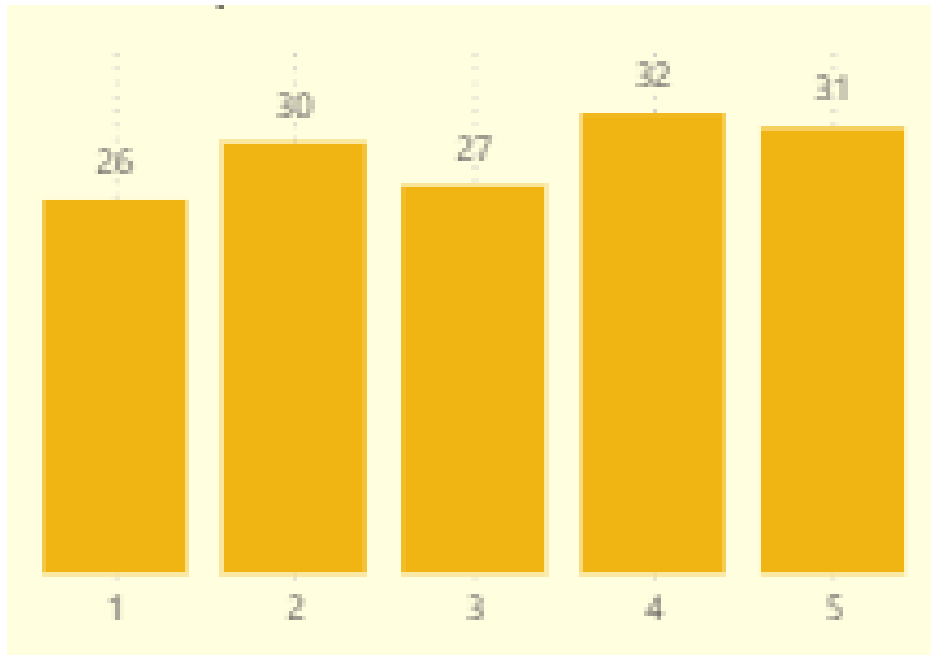


Figure 6.5: Churn by Satisfaction Rating

Business Insight

The dashboard demonstrates a strong relationship between customer satisfaction and churn behavior. Customers reporting lower satisfaction scores contribute disproportionately to customer attrition, confirming that customer experience remains one of the strongest predictors of retention.

This finding reinforces a key Business Intelligence principle: customer satisfaction is a leading indicator of future revenue performance.

Business Implication

Declining customer satisfaction today is likely to translate into increased churn and reduced revenue tomorrow.

Churn Rate by State

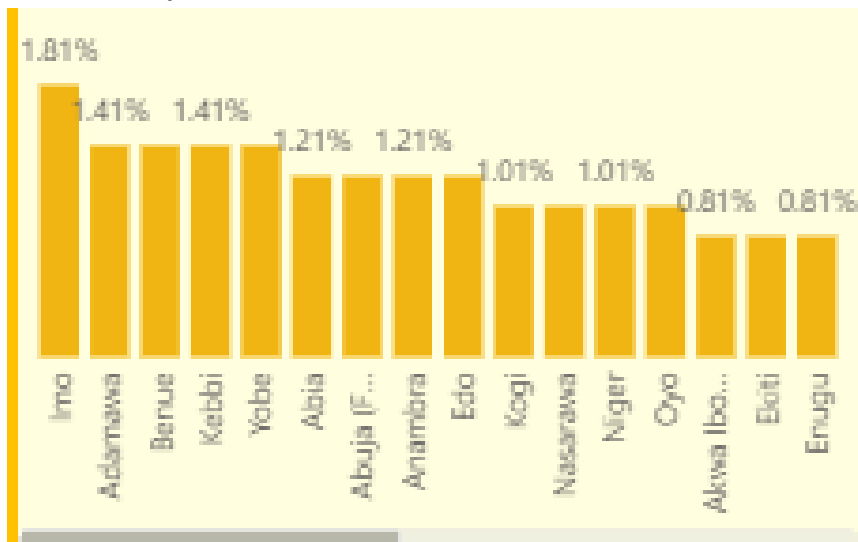


Figure 6.6: Churn Rate by State

Business Insight

Regional analysis indicates that churn is concentrated within selected states, suggesting that customer attrition is influenced by localized market conditions rather than national performance alone. This insight enables MTN to replace expensive nationwide retention initiatives with targeted regional interventions.

CHAPTER SEVEN

Customer Retention and Risk Analysis

Customer retention represents the ultimate objective of Business Intelligence within the telecommunications industry. While previous chapters identified the causes of customer attrition, this chapter focuses on protecting valuable customer relationships through proactive risk management and retention strategies.

The Customer Retention and Risk Dashboard provide executives with predictive insights into customer risk, revenue exposure, customer lifetime value, and retention performance. Rather than reacting to churn after customers leave, the dashboard supports evidence-based decision-making that enables MTN to intervene before attrition occurs.

Customer Retention and Risk Dashboard

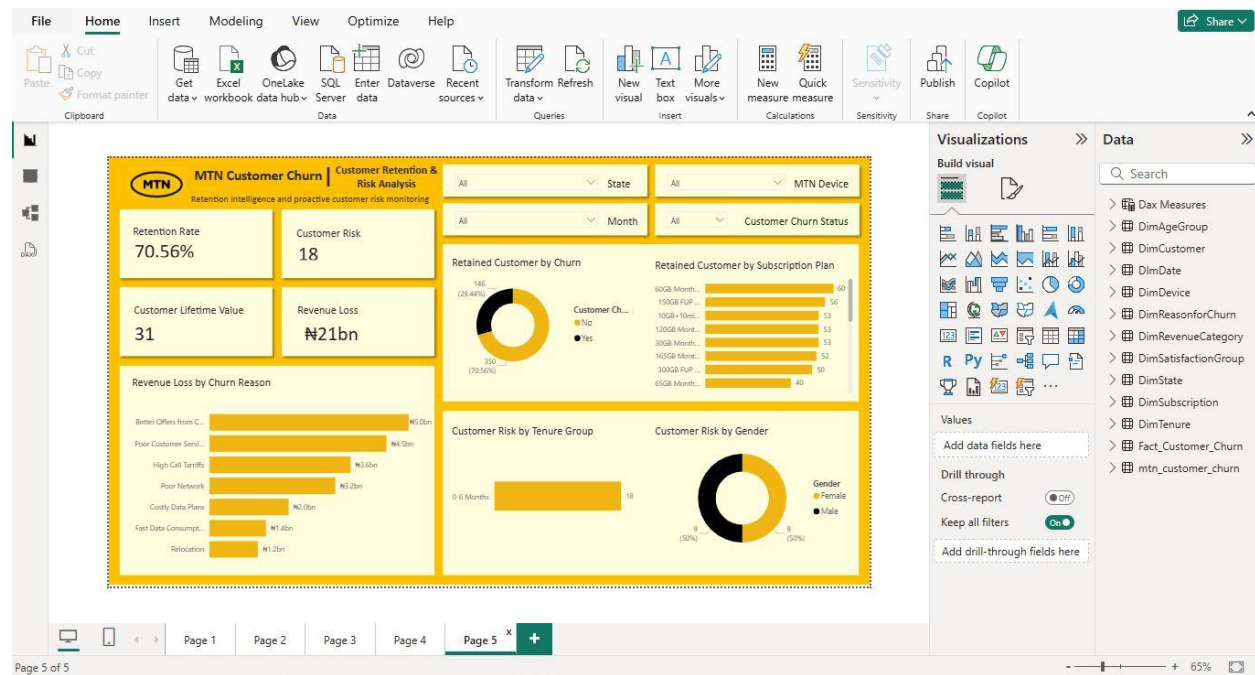
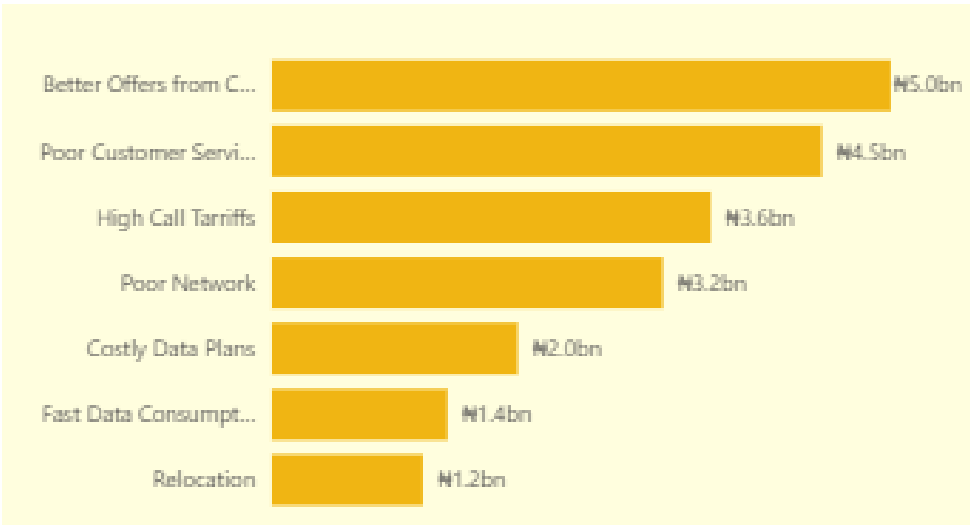


Figure 7.1: Customer Retention and Risk Dashboard

Revenue Loss Analysis



The dashboard estimates that customer churn has resulted in approximately ₦21 billion in lost revenue, demonstrating the substantial financial consequences of customer attrition. Revenue loss is highest among customers citing better competitor offers, poor customer service, high call tariffs, and network-related issues.

From an executive perspective, this finding confirms that churn is not merely a customer issue but a direct threat to financial performance and shareholder value.

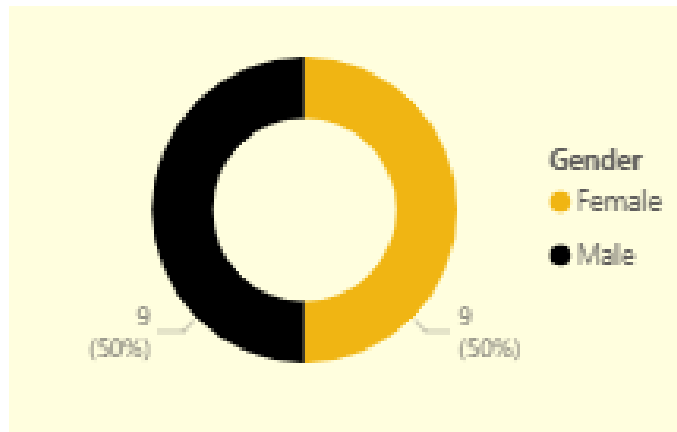
Customer Risk Analysis



Customer Risk by Tenure Group

Business insight: The dashboard identifies customers within the 0–6 months tenure group as the highest-risk segment. This suggests that the earliest stage of the customer lifecycle represents the period during which loyalty has not yet been established.

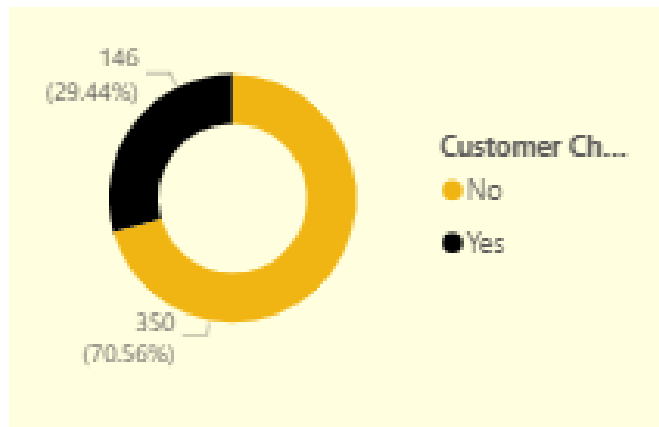
Customer Risk by Gender



Customer Risk by Gender

Business insight: The analysis indicates that customer risk is distributed relatively evenly across male and female customers. This finding suggests that gender is not a primary determinant of churn behavior.

Retained Customers by Subscription Plan



Retained Customers by Subscription Plan

Business insight: This visualization highlights the subscription plans that retain the largest number of customers. Products with consistently high retention demonstrate strong customer value and should serve as benchmarks for future product development.

Summary of the Study

The Business Intelligence analysis produced several strategic findings that significantly influence MTN's commercial performance.

The Executive Dashboard revealed that while MTN generated approximately ₦68.75 billion in revenue from 496 customers, the organization experienced a customer churn rate of 29.44%, representing a considerable threat to recurring revenue and future profitability. Although customer acquisition remains strong, customer retention emerged as the more significant determinant of long-term financial sustainability.

Customer Analysis further demonstrated that customer behavior differs across age groups, tenure categories, geographical regions, and device preferences. These behavioral differences directly influence purchasing decisions, customer satisfaction, and Customer Lifetime Value (CLV), emphasizing the need for personalized customer engagement strategies.

Revenue Analysis showed that broadband products and premium subscription plans contribute disproportionately to total revenue, indicating significant opportunities to increase profitability through targeted product optimization and value-based customer segmentation.

The Churn Deep Dive identified poor customer service, competitor offerings, network-related concerns, and pricing dissatisfaction as the primary drivers of customer attrition. The analysis further confirmed that customer churn is concentrated within specific subscription plans and geographical regions rather than being evenly distributed across the customer base.

Finally, the Customer Retention and Risk Dashboard demonstrated that early-tenure customers represent the highest retention risk while also quantifying the substantial financial losses associated with customer churn. These findings reinforce the importance of predictive analytics and proactive customer engagement in reducing customer attrition.

Collectively, the analysis confirms that sustainable business growth depends not only on acquiring new customers but also on maximizing the value of existing customer relationships through improved customer experience, operational excellence, and data-driven decision-making.

Recommendations

Improve Customer Retention Through Predictive Analytics

One of the most significant findings of this project is the customer churn rate of 29.44%, indicating that approximately three out of every ten customers discontinue MTN services during the reporting period. This represents a substantial loss of recurring revenue and customer lifetime value. To address this challenge, MTN should implement predictive churn analytics capable of identifying customers at risk before they leave. Customer behavior such as declining purchase frequency, reduced data usage, repeated complaints, declining satisfaction scores, and shortened engagement periods should be monitored continuously to generate churn risk alerts.

Proactive retention campaigns—including personalized offers, loyalty rewards, and targeted customer engagement—should then be initiated before customers decide to migrate to competing providers.

Expected Business Benefits

- * Reduction in customer churn.
- * Increased Customer Lifetime Value (CLV).
- * Lower customer acquisition costs.
- * Improved recurring revenue.

Enhance Customer Experience

Customer satisfaction analysis demonstrated a strong relationship between customer experience and churn behavior. Customers reporting lower satisfaction scores were significantly more likely to discontinue MTN services. Executive management should therefore prioritize continuous improvements in customer service delivery, complaint resolution processes, network reliability, and digital customer support.

Customer feedback should be monitored in real time using Business Intelligence dashboards to identify emerging service issues before they affect customer loyalty.

Expected Business Benefits

- * Higher customer satisfaction.
- * Increased customer loyalty.
- * Reduced complaints.
- * Improved brand reputation.

Optimize Subscription Plans

The analysis revealed that churn is concentrated within selected subscription plans, suggesting that certain products no longer meet customer expectations regarding pricing, value, or service quality. MTN should periodically evaluate the performance of all subscription plans using customer adoption rates, revenue contribution, profitability, and churn performance. Subscription plans experiencing persistent customer losses should be redesigned through improved pricing structures, increased data allocations, or additional value-added services.

Expected Business Benefits

- * Improved product competitiveness.
- * Increased customer retention.
- * Higher Average Revenue per User (ARPU).
- * Stronger product adoption.

Strengthen Regional Market Strategy

Customer distribution and churn performance varied considerably across states, indicating that commercial opportunities differ significantly between geographical markets. Rather than applying identical commercial strategies nationwide, MTN should adopt state-specific marketing initiatives supported by Business Intelligence.

Network expansion, promotional campaigns, and customer engagement activities should be prioritized according to customer value, market potential, revenue contribution, and churn performance within each region.

Expected Business Benefits

- * Better resource allocation.
- * Increased regional market penetration.
- * Higher return on investment.
- * Improved competitive positioning.

Increase Investment in Broadband Services

Revenue analysis demonstrated that broadband-enabled products contribute significantly to organizational revenue. As customer demand continues shifting toward high-speed internet connectivity, MTN should continue expanding broadband infrastructure while introducing affordable broadband packages capable of attracting both residential and business customers. Bundled services combining voice, internet, and digital applications should also be introduced to strengthen customer loyalty.

Expected Business Benefits

- * Increased revenue.
- * Higher broadband adoption.
- * Improved customer value.
- * Stronger competitive advantage.

Implement Customer Lifecycle Management

The dashboard analysis identified customers within the first six months of subscription as the highest-risk customer segment. MTN should therefore establish a structured Customer Lifecycle Management framework that supports customers throughout their relationship with the organization. Personalized onboarding programs, educational campaigns, proactive customer engagement, and milestone-based loyalty rewards should be introduced to improve long-term retention.

Expected Business Benefits

- * Reduced early customer churn.
- * Increased customer engagement.
- * Improved retention.
- * Higher Customer Lifetime Value.

Strengthen Business Intelligence Capability

Business Intelligence should become an integral component of executive decision-making rather than a reporting function. MTN should establish an enterprise-wide Business Intelligence Centre responsible for predictive analytics, dashboard management, performance monitoring, executive reporting, and data governance. This will enable management to make faster, evidence-based decisions while continuously monitoring customer behavior, revenue performance, and operational efficiency.

Expected Business Benefits

- * Faster decision-making.
- * Improved operational performance.
- * Enhanced strategic planning.
- * Sustainable competitive advantage.

Conclusion

This project demonstrates the practical application of Business Intelligence techniques in addressing one of the telecommunications industry's most significant challenges: customer churn. By integrating data modelling, Power BI dashboard development, KPI design, and executive-level analytics, the study transformed raw customer data into meaningful business intelligence capable of supporting strategic decision-making.

The dashboards developed throughout this project provided comprehensive visibility into MTN's financial performance, customer behavior, revenue generation, customer retention, and operational risks. More importantly, the project demonstrated how analytical insights can be translated into actionable recommendations that improve organizational performance and strengthen long-term competitiveness.

The findings confirm that customer retention, customer experience, and customer lifetime value represent the primary drivers of sustainable profitability within the telecommunications industry. Organizations capable of anticipating customer behavior, personalizing customer engagement, and leveraging predictive analytics will achieve superior financial performance and stronger competitive advantage.

From a professional perspective, this portfolio illustrates the complete Business Intelligence lifecycle, beginning with business problem identification, progressing through data preparation, modelling, dashboard development, analytical interpretation, and concluding with strategic recommendations for executive management. The project therefore demonstrates not only technical proficiency in Microsoft Power BI but also the ability to translate data into business value through analytical thinking and evidence-based decision-making.

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